

Supplementary material

Urban boundaries and gender gaps in stochastic accessibility:

Evidence from Bogotá

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1 Supplementary methods

1.1 Interpretation of the stochastic-accessibility estimand

The origin–destination Markov chain represents aggregate survey-weighted population flows. A first-passage step is a dimensionless transition in this fitted flow system and must not be interpreted as a minute of travel or as an observed individual’s itinerary. For every purpose, residential distributions are conditioned on starting outside the corresponding target set in the primary specification.

1.2 Regularization sensitivity

Group-specific rows are shrunk toward the population kernel using $\omega_{ig} = \text{ESS}_{ig} / (\text{ESS}_{ig} + \tau)$. The primary analysis uses $\tau = 20$. We re-estimated both territorial systems at $\tau \in \{0, 10, 20, 50\}$. At $\tau = 0$, origins with positive group-specific support use the empirical row, whereas origins without such support retain the population row so that the transition kernel remains defined. Table S1 reports the total gap, its paired scope contrast, the corresponding residential and kernel contrasts, and the normalized-gap contrast. It also retains the origin-level magnitude and rank diagnostics relative to $\tau = 20$. These are point-estimate sensitivity calculations; inferential intervals in the main analysis correspond to the primary $\tau = 20$ specification.

Table S1: Sensitivity to the row-shrinkage parameter. Panel A reports the main point estimands after complete re-estimation at each value of τ . G_D and G_R are the female-minus-male gaps for Bogotá D.C. and Bogotá-Region; $\Delta G = G_R - G_D$, with analogous paired contrasts for the residential (ΔR) and kernel (ΔK) components. ΔG_{rel} is expressed in percentage points. Panel B reports origin-level FPT changes and rank agreement relative to the primary $\tau = 20$ specification.

Panel A: scope-gap estimands

τ	Purpose	G_D	G_R	ΔG	ΔR	ΔK	ΔG_{rel} (pp)
0	Employment	0.521	1.078	0.557	-0.022	0.579	9.81
0	Education	0.440	0.908	0.468	-0.023	0.491	8.16
0	Health	0.256	0.753	0.497	-0.030	0.527	6.43
10	Employment	0.499	1.037	0.538	-0.022	0.560	9.52
10	Education	0.418	0.866	0.448	-0.023	0.470	7.84
10	Health	0.249	0.729	0.481	-0.030	0.510	6.21
20	Employment	0.479	0.999	0.520	-0.022	0.542	9.24
20	Education	0.399	0.827	0.429	-0.022	0.451	7.54
20	Health	0.240	0.705	0.465	-0.030	0.495	6.02
50	Employment	0.428	0.900	0.472	-0.022	0.494	8.47
50	Education	0.350	0.730	0.380	-0.022	0.402	6.75
50	Health	0.218	0.641	0.423	-0.029	0.453	5.50

Panel B: origin-level FPT diagnostics relative to $\tau = 20$

τ	Max. absolute change	Min. Spearman	Median Spearman
10	0.078	0.949	0.989
20	0.000	1.000	1.000
50	0.103	0.971	0.990

1.3 Specification grid

The effective robustness grid crosses two target definitions (core 50% and extended 80% of supported weighted arrivals) and two residential conventions (outside-target conditioning or all residences) across three purposes, yielding 12 specifications. The reference kernel uses the complete survey population. Restricting its construction to records with valid sex categories changes the displayed estimates by less than the reported precision. Tables S2 and S3 report the 12 distinct full-population specifications.

Table S2: Range of regional female-minus-male decomposition estimates over the 12 effective full-population robustness specifications.

Purpose	Total gap range	Residential range	Kernel range
Education	[0.090, 0.827]	[-0.013, 0.030]	[0.077, 0.840]
Employment	[0.127, 0.999]	[-0.043, -0.020]	[0.147, 1.031]
Health	[0.493, 0.959]	[-0.046, -0.031]	[0.532, 0.997]

Table S3: Regional female-minus-male decomposition for the 12 effective full-population robustness specifications.

Target	Conditioning	Purpose	Total gap	Residential	Kernel
Core 50%	Outside	Employment	0.999	-0.033	1.031
Core 50%	All	Employment	0.754	-0.043	0.797
Extended 80%	Outside	Employment	0.358	-0.020	0.378
Extended 80%	All	Employment	0.127	-0.020	0.147
Core 50%	Outside	Education	0.827	-0.013	0.840
Core 50%	All	Education	0.592	0.030	0.562
Extended 80%	Outside	Education	0.239	-0.012	0.251
Extended 80%	All	Education	0.090	0.013	0.077
Core 50%	Outside	Health	0.705	-0.031	0.736
Core 50%	All	Health	0.570	-0.046	0.616
Extended 80%	Outside	Health	0.959	-0.038	0.997
Extended 80%	All	Health	0.493	-0.039	0.532

1.4 Multiscale territorial construction

External municipalities are ranked by total bidirectional survey-weighted flow with Bogotá using the complete population. Nested functional scopes include the minimum ranked municipalities required to reach at least 50% and 80% of that interface flow. Together with Bogotá D.C. and the complete regional network, the realized interface shares are 0, 0.564, 0.824, and 1. Within every scope, the largest directed strongly connected component, population kernel, purpose targets, group kernels,

Table S4: Final Bogotá-Region household-bootstrap inference.

Purpose	Effect	Estimate	SD	Pointwise 95% CI	Simultaneous 95% CI
Education	Total gap	0.827	0.168	[0.547, 1.194]	[0.451, 1.204]
Education	Residential	-0.013	0.024	[-0.054, 0.037]	[-0.067, 0.042]
Education	Mobility kernel	0.840	0.165	[0.581, 1.206]	[0.467, 1.213]
Employment	Total gap	0.999	0.164	[0.718, 1.370]	[0.632, 1.366]
Employment	Residential	-0.033	0.024	[-0.074, 0.018]	[-0.088, 0.023]
Employment	Mobility kernel	1.031	0.160	[0.758, 1.375]	[0.671, 1.392]
Health	Total gap	0.705	0.203	[0.363, 1.109]	[0.252, 1.159]
Health	Residential	-0.031	0.029	[-0.085, 0.028]	[-0.096, 0.035]
Health	Mobility kernel	0.736	0.196	[0.411, 1.133]	[0.294, 1.178]

and residential distributions are reconstructed. The full-region endpoint retains all 141 nodes in the validated regional kernel, including one node without a municipality label.

1.5 Bootstrap inference

The final regional analysis uses 500 joint Poisson(1) household-cluster bootstrap replicates. The same multiplier is assigned to every record from a household and is propagated jointly to person and trip weights. Simultaneous intervals use a bootstrap max- t critical value over the three destination purposes within each effect family. The paired scope analysis likewise uses 500 shared household multipliers in the regional and district systems. The earlier 50-replicate specification grid and 200-replicate standalone district analysis are robustness diagnostics rather than the final paired inference.

2 Supplementary results

2.1 Regional gap estimates

Table S4 reports the regional gaps and their household-bootstrap intervals for each purpose.

2.2 District support and target reconstruction

The Bogotá D.C.-internal analysis retains the largest directed strongly connected component. Purpose-specific targets are reconstructed within the district scope rather than inherited from the regional system.

Table S5: Support and strongly connected component diagnostics for the Bogotá D.C.-internal network.

Metric	Value
City utam in zoning	116.000
Largest scc nodes	114.000
Internal trip rows before scc	76168.000
Internal trip rows in scc	76168.000
Trip weight scc coverage	1.000
City person rows before scc	48989.000
City person rows in scc	48373.000
Person weight scc coverage	0.998

Table S6: Scope-specific Bogotá D.C.-internal opportunity target sets.

Purpose	Definition	Target nodes	Supported arrival share
Employment	core_50	24	0.500
Education	core_50	24	0.512
Health	core_50	17	0.508

2.3 Paired territorial-scope contrasts

Positive step-valued contrasts indicate a larger female-minus-male gap in the regional system. Relative attenuation is expressed as a percentage of the regional gap, as reported in Table S7. The scope contrast combines changes in the network, opportunity geography, and residential population.

2.4 Multiscale boundary response

The total female-minus-male gap is regressed on realized interface-flow coverage within each bootstrap replicate. Max- t simultaneous intervals are formed across the three purposes separately for each decomposition component. Table S8 reports the fitted slopes. Positive simultaneous intervals support an overall increasing response, but do not imply that every successive scope estimate is larger.

Point-estimate monotonicity holds for employment but not for education or health. The corresponding bootstrap probabilities of a nondecreasing total-gap sequence are 0.756, 0.110, and 0.212. The main inference is therefore the positive fitted response, not strict local monotonicity.

Table S7: Paired Bogotá-Region minus Bogotá D.C.-internal scope contrasts.

Purpose	Effect	Unit	Estimate	SD	Simultaneous 95% CI
Employment	Total gap	FPT steps	0.520	0.153	[0.178, 0.861]
Employment	Residential	FPT steps	-0.022	0.025	[-0.078, 0.034]
Employment	Mobility kernel	FPT steps	0.542	0.146	[0.217, 0.866]
Employment	Relative attenuation	Percent	52.035	8.777	[31.831, 72.240]
Education	Total gap	FPT steps	0.429	0.160	[0.070, 0.787]
Education	Residential	FPT steps	-0.022	0.024	[-0.076, 0.031]
Education	Mobility kernel	FPT steps	0.451	0.155	[0.108, 0.794]
Education	Relative attenuation	Percent	51.821	11.178	[26.090, 77.552]
Health	Total gap	FPT steps	0.465	0.169	[0.087, 0.843]
Health	Residential	FPT steps	-0.030	0.028	[-0.091, 0.032]
Health	Mobility kernel	FPT steps	0.495	0.162	[0.136, 0.853]
Health	Relative attenuation	Percent	65.911	15.801	[29.537, 102.284]

Table S8: Multiscale boundary-response slopes per unit of Bogotá-external interface-flow coverage.

Purpose	Total gap	Residential	Mobility kernel
Employment	0.451 [0.161, 0.741]	-0.018 [-0.058, 0.023]	0.469 [0.175, 0.762]
Education	0.320 [0.024, 0.616]	-0.018 [-0.057, 0.021]	0.338 [0.035, 0.641]
Health	0.391 [0.052, 0.730]	-0.026 [-0.073, 0.022]	0.417 [0.078, 0.755]

2.5 Socioeconomic-stratum diagnostic

The stratum diagnostic uses six ordered socioeconomic strata. Consequently, the Spearman statistic has discrete support, and percentile interval endpoints may coincide with the observed coefficient when estimates approach -1 or 1 .

2.6 Paired specification-grid sensitivity

The regional-minus-district contrast was recalculated for the same 12 effective target-definition and conditioning combinations. All total-gap, kernel-component, and relative-gap contrasts remained positive. Magnitudes were nevertheless specification-dependent: the smallest employment and

Table S9: Association between socioeconomic stratum and observed mean FPT.

Purpose	Spearman rho	Bootstrap SD	Lower 95% CI	Upper 95% CI
Education	0.886	0.141	0.314	0.886
Employment	-1.000	0.046	-1.000	-0.829
Health	-0.771	0.120	-0.829	-0.429

education contrasts occurred when the extended 80% target was combined with all residences. These are point-estimate diagnostics and do not inherit the simultaneous confidence intervals of the primary specification.

Table S10: Paired specification-grid sensitivity. G_D and G_R are the female-minus-male FPT gaps for Bogotá D.C. and Bogotá-Region. The remaining columns report regional-minus-district contrasts for the total gap, residential component, kernel component, and relative gap in percentage points. All entries are point estimates.

Target	Conditioning	Purpose	G_D	G_R	ΔG	ΔR	ΔK	ΔG_{rel} (pp)
Core 50%	Outside	Employment	0.479	0.999	0.520	-0.022	0.542	9.24
Core 50%	All	Employment	0.355	0.754	0.399	-0.017	0.416	9.33
Extended 80%	Outside	Employment	0.269	0.358	0.090	0.002	0.088	3.62
Extended 80%	All	Employment	0.094	0.127	0.033	0.004	0.029	3.93
Core 50%	Outside	Education	0.399	0.827	0.429	-0.022	0.451	7.54
Core 50%	All	Education	0.302	0.592	0.290	-0.019	0.309	7.12
Extended 80%	Outside	Education	0.117	0.239	0.121	-0.009	0.131	4.88
Extended 80%	All	Education	0.056	0.090	0.034	-0.004	0.038	4.51
Core 50%	Outside	Health	0.240	0.705	0.465	-0.030	0.495	6.02
Core 50%	All	Health	0.189	0.570	0.381	-0.037	0.418	5.84
Extended 80%	Outside	Health	0.259	0.959	0.699	-0.024	0.723	15.66
Extended 80%	All	Health	0.126	0.493	0.368	-0.016	0.384	15.81

2.7 Full municipal omission and sample support

As a sensitivity diagnostic distinct from fixed-kernel origin attribution, each of the 20 external municipalities is omitted in turn. We remove its UTAMs, all incident survey trips, and its residents; then reconstruct the largest directed strongly connected component, population kernel, sex-specific kernels, residential distributions, and purpose-specific core-50 targets. The Bogotá D.C. reference remains fixed. A regional UTAM without a municipality label is retained in all fits. All 20 refits and the unmodified 141-node regional baseline pass the reconstruction checks. Facatativá and Soacha have respectively 229 and 1,449 sampled outbound households; their female/male outbound trip effective sample sizes are 299/241 and 2,246/1,944. These support statistics describe the available sample and do not quantify uncertainty in the refitted contrasts. Table S11 gives the ranges and the two most relevant named examples. This diagnostic redefines the network and target geography and is neither a paired bootstrap test nor a causal deletion effect.

Table S11: Full leave-one-external-municipality-out sensitivity at the primary specification. For each omitted municipality, the directed strongly connected component, population and sex-specific kernels, residential distributions, and core-50 targets are re-estimated. The contrast compares the re-estimated regional female-minus-male gap with the unchanged Bogotá D.C. estimate. All entries are descriptive point estimates in FPT steps; the baseline contrasts are 0.520, 0.429, and 0.465 for employment, education, and health.

Purpose	Omitted municipality	Refit contrast	Change from baseline
Employment	Facatativá	0.277	-0.243
Education	Facatativá	0.080	-0.348
Health	Facatativá	0.270	-0.195
Employment	Soacha	0.389	-0.131
Education	Soacha	0.560	+0.131
Health	Soacha	0.347	-0.118

Across all 20 omissions, the refit contrast ranges are [0.277, 0.565] for employment, [0.080, 0.564] for education, and [0.270, 0.552] for health. The corresponding minimum differences of normalized regional and district gaps are 0.052, 0.010, and 0.036. Facatativá has 229 sampled outbound households, 635 outbound trips, and 586 sampled residents in the regional network; Soacha has 1,449 households, 4,728 outbound trips, and 4,496 sampled residents. Each omission changes the regional model and the opportunity set; these differences are not causal effects.

2.8 Time-threshold benchmark and home-origin kernels

A fixed-time comparator was built on the same directed UTAM networks and scope-specific core-50 purpose sets as the primary analysis. Survey-weighted mean positive reported minutes provide edge costs; shortest directed travel-time paths determine which destinations are reachable within 45 or 60 minutes. Each destination’s weight is its share of supported pooled weighted arrivals for the purpose and scope. For each sex, the score is averaged over the original survey-weighted residential distribution conditional on starting outside the target set. The female disadvantage is defined as male minus female reachable-arrival share. Its paired regional contrast uses the same fixed time surfaces for women and men within each scope. A 500-replicate shared Poisson household bootstrap reweights only the residential distributions, conditional on the estimated edge costs and targets; Table S12 gives unadjusted percentile intervals. This is a conventional time-threshold *proxy* based on revealed OD edges and arrivals, not an inventory of opportunities or a complete route-choice model. It lacks a sex-specific movement kernel by construction and cannot be compared numerically to FPT steps.

A separate check identifies home-origin trips when a trip starts in its household’s residence UTAM. The full-population reference kernel, connected network, destinations, and residential distributions are held fixed; only the sex-specific empirical trip rows and their effective sample sizes change. All

trip records were assigned to households with a unique residential UTAM. The pooled weighted share of home-origin trips was 0.623. Table S12 also reports the resulting FPT point contrasts. They do not inherit intervals from the primary model.

Table S12: Deterministic time-access benchmark and home-origin kernel sensitivity.

Panel A. Time-threshold benchmark.

Purpose	Time threshold	Regional-minus-district access gap	95% interval
Employment	45 minutes	-0.0008	[-0.0036, 0.0023]
Employment	60 minutes	-0.0017	[-0.0051, 0.0021]
Education	45 minutes	-0.0028	[-0.0054, -0.0001]
Education	60 minutes	-0.0021	[-0.0053, 0.0013]
Health	45 minutes	-0.0014	[-0.0042, 0.0014]
Health	60 minutes	-0.0017	[-0.0056, 0.0018]

Panel A: the benchmark gap is men’s minus women’s residentially weighted share of supported core-50 target-arrival weight reachable within the stated threshold. A positive regional-minus-district contrast would indicate greater disadvantage for women in the regional time-access proxy. Intervals are percentile intervals from 500 shared Poisson household resamples, conditional on fixed destinations and pooled time surfaces. They are descriptive and unadjusted across six comparisons. Reported edge durations cover all survey-weighted trips in the validated regional network. The pooled time surface has no group-specific mobility kernel.

Panel B. Home-origin sex-kernel sensitivity of the paired FPT contrast.

Purpose	Primary contrast	Home-origin contrast	Ratio
Employment	0.5197	0.5579	1.073
Education	0.4287	0.5044	1.176
Health	0.4649	0.3331	0.717

Contrasts are regional-minus-district female-minus-male mean FPT in transitions. The home-origin diagnostic restricts empirical group kernel rows to trips starting in each household’s residential UTAM, while keeping the full-population reference kernel, original residence distributions, scope-specific targets, and network components fixed. The full-region home-origin sex kernels draw on 32,504 sampled trips for women and 28,409 for men; the district sex kernels draw on 23,969 and 20,702, respectively. These are point estimates without a dedicated uncertainty interval.

2.9 Paired bootstrap with reselected targets

In 150 exploratory shared Poisson household replicates, we compared the regional-minus-district FPT contrast under the fixed primary target sets and under newly selected supported core-50 targets inside each replicate. The original strongly connected components and population shrinkage kernels were retained; sampled residential masses and sex-specific transition row weights and effective sample sizes varied in both variants. Table S13 gives percentile intervals from identical resamples, allowing a direct comparison of widths. They are not simultaneous intervals, and the smaller run does not replace the primary 500-replicate max- t inference.

Table S13: Reduced paired-bootstrap sensitivity to reselecting purpose targets.

Purpose	Fixed target interval	Reselected interval	Fixed width	Reselected width
Employment	[0.287, 0.953]	[0.184, 0.799]	0.666	0.615
Education	[0.179, 0.849]	[0.050, 0.904]	0.670	0.855
Health	[0.107, 0.910]	[0.048, 0.928]	0.803	0.880

Exploratory percentile intervals for the regional-minus-district female-minus-male mean FPT contrast in 150 paired Poisson household replicates, conditional on the original network components and full-sample population kernel. The identical resamples are used in both columns. In the second column supported core-50 target sets are reconstructed inside each replicate; sex-specific row ESS and residential masses are reweighted in both columns. These unadjusted percentile intervals do not replace the primary 500-replicate simultaneous max- t intervals. Target Jaccard median across replicates: district/region 0.846/0.844 for employment, 0.778/0.758 for education, and 0.750/0.810 for health.

2.10 Network, target, and resident-population accounting

Let G_D denote the district gap among residents of the 114 district UTAMs in the largest strongly connected component, evaluated on the district network and its core-50 destination set. Let G_{RC} use the regional transition kernels with the same residents and district destinations, G_{RT} use regional kernels and destinations with the same residents, and G_R include all eligible regional residents and regional destinations. Within each state, women’s and men’s residential weights are normalized after excluding the respective destination set. The ordered identity is

$$G_R - G_D = (G_{RC} - G_D) + (G_{RT} - G_{RC}) + (G_R - G_{RT}). \quad (1)$$

The first term retains the district’s starting population and targets while changing the fitted transition system; the second changes the target geography; the third expands the residential population. The decomposition is an accounting identity for this order of comparisons. Its terms do not represent independent causal effects, and a different order could allocate interactions differently.

The same Poisson(1) household multiplier is used in all four states within each of 500 replicates. The network components, population reference kernels, and core-50 destination sets are fixed at their full-sample definitions. Max- t intervals are simultaneous across the three purposes for each term separately. The original district and regional point estimates are recovered, and the three contributions sum to their regional-minus-district difference in every replicate. Table S14 reports the four contrasts.

Table S14: Ordered accounting of the regional-minus-district female-minus-male first-passage gap. Values are FPT transitions. Intervals are 95% max- t simultaneous intervals over the three destination purposes within each row; interval families are separate across rows. The full regional and district point estimates reproduce the primary estimates, whereas these intervals arise from a separate 500-replicate household bootstrap and do not replace primary Table S7.

Ordered contrast	Employment	Education	Health
Regional network, city residents and city targets minus district	0.015 [-0.051, 0.082]	0.052 [-0.031, 0.136]	-0.027 [-0.119, 0.065]
Regional targets minus city targets, city residents	0.002 [-0.102, 0.106]	-0.056 [-0.195, 0.083]	0.001 [-0.109, 0.110]
All regional residents minus city residents, regional targets	0.502 [0.156, 0.848]	0.433 [0.090, 0.775]	0.491 [0.138, 0.845]
Regional minus district total	0.520 [0.135, 0.904]	0.429 [0.050, 0.808]	0.465 [0.050, 0.879]

3 Reproducibility and interpretation boundaries

Tables S1–S11 report the primary analyses and model sensitivities; Tables S12–S14 report the time-access benchmark, target-selection sensitivity, and ordered territorial accounting. Analysis code and input definitions are documented in the repository. The analysis is cross-sectional and descriptive. Survey weights and clustered bootstrap inference propagate sampling uncertainty but do not remove recall error, nonresponse bias, unobserved modal and temporal heterogeneity, or the limited gender categories in the processed survey.